Problem Statement

Inconsistent treatment and **inefficient use of hospital resources** arise from the current skeletal traction device:

1. The **traction frame is fixed** to a hospital bed, which impacts the ability to perform necessary scans to develop accurate treatment plans. It also requires a special hospital bed, which is expensive and can take time to bring to the patient.
2. **Hanging weights** (around 20 pounds) do not apply consistent traction, introduce risk of injury to surrounding people, and requires a staff member to hold the weight during patient transport.
3. The geometry of the solution causes the **patient to slide down the bed**, which also leads to inadequate traction and need for staff to frequently readjust the patient.

Needs Statement

Orthopedic surgeons need a way to apply skeletal traction to **adult patients with femoral shaft fractures** that **maintains correct femur length** throughout treatment despite changes in patient position to **prevent femur displacement** that can cause further damage.

User Needs

1. Device shall apply traction to femur fracture per current treatment requirements	7. Device shall provide line of sight and access to the pin site
2. The device shall maintain the set length and position of femur throughout treatment	8. Device shall be biocompatible
3. User must be able to know the force applied	9. Device shall be sterilizable
4. Device shall be radiolucent for x-rays	10. Device packaging, and labeling shall be consistent with FDA standards
5. The device must be compatible with use of most hospital bed, including X-ray beds	11. Device shall be composed of reusable components
6. The pin must align with current traction pin use and safety standards for distal femoral pins or proximal tibial pins	